Django Final Projects

Project Overview

For your final project, you will build a **modern and responsive Django website** using:

- **Django** — Backend
- **HTML** — Website structure
- **CSS** — Design
- **JavaScript** — Interactivity
- **Database** — Store and manage your data

Important Requirement

Every project must have at least 3 Django models.

Your website must be **dynamic**, meaning that the information displayed on the website should come from the database.

Project 1 — E-Commerce Website

Idea

Build an online store where users can:

Browse Products → Search → View Details → Add to Cart → Checkout → Place an Order

Home Page

Include:

- Store name/logo
- Navigation bar
- About the store
- Hero section
- Search bar
- Product categories
- Featured products

Products and Cart

Users must be able to:

- View all products
- Search products
- Filter by category
- View product details
- See name, description, price, image, and rating/review
- Add products to cart
- Remove products from cart
- Increase/decrease quantity
- See the total price

Checkout

The customer must enter:

- Name **(required)**
- Phone number
- Address

Then:

1. Show the order summary.
2. Show the total price.
3. Confirm the order.
4. Save the order in the database.
5. Show a success message.

Example:

> Thank you, your name! Your order has been placed successfully.

Minimum Models

You need **at least 3 models**.

Product

- name
- description
- price
- image
- category
- rating

Order

- Product name
- customer_name
- phone
- address
- total_price
- created_at

OrderItem

- product
- quantity
- price

Project 2 — Restaurant Ordering Website

Idea

Build a restaurant website where customers can:

Browse Meals → Search → View Details → Add to Cart → Checkout → Place an Order

Home Page

Include:

- Restaurant name/logo
- Navigation bar
- Restaurant information
- Hero section
- Search bar
- Food categories
- Popular meals

Menu and Cart

Customers must be able to:

- View all meals
- Search meals
- Filter by category
- View meal details
- See name, description, price, image, and rating/review
- Add meals to cart
- Remove meals from cart
- Increase/decrease quantity
- See the total price

Checkout

The customer must enter:

- Name **(required)**
- Phone number
- Address **or table number**

Then:

1. Show the order summary.
2. Show the total.
3. Confirm the order.
4. Save the order in the database.
5. Show a thank-you message.

Example:

> Thank you, your name! Your order has been placed successfully.

Minimum Models

You need **at least 3 models**.

MenuItem

- name
- description
- price
- image
- category
- rating

Order

- customer_name
- phone
- address
- total_price
- created_at

OrderItem

- menu_item
- quantity
- price

Project 3 — Online Bookstore

Idea

Build an online bookstore where users can:

Browse Books → Search → View Details → Add to Cart → Checkout → Buy Books

Home Page

Include:

- Store/library name and logo
- Navigation bar
- About the bookstore
- Hero section
- Search bar
- Book categories
- Featured books
- Popular authors

Books and Cart

Users must be able to:

- View all books
- Search books
- Filter by category
- View book details
- See:
    - Name
    - Author
    - Description
    - Price
    - Image
    - Rating/review
    - Publication year
- Add books to cart
- Remove books from cart
- Increase/decrease quantity
- See the total price

Checkout

The customer must enter:

- Name **(required)**
- Phone number
- Address

Then:

1. Show the order summary.
2. Show the total.
3. Confirm the purchase.
4. Save the order in the database.
5. Show a success message.

Example:

> Thank you, your name! Your order has been placed successfully.

Minimum Models

You need **at least 3 models**.

Book

- name
- author
- description
- price
- image
- category
- rating
- publication_year

Order

- customer_name
- phone
- address
- total_price
- created_at

OrderItem

- book
- quantity
- price

Design Requirements

Your website must look **modern, beautiful, and responsive**.

Must Include

- Navigation bar
- Hero section
- Cards
- Modern buttons
- Good colors
- Good spacing
- Hover effects
- Simple animations
- Mobile-friendly design
- Responsive layout

Do Not Submit

- Plain HTML pages
- Default browser design
- Unstyled forms
- Pages with no visual design

JavaScript Requirements

You must use **JavaScript** to make your website interactive.

Must Include

- Search
- Filtering

- Add to cart
- Remove from cart
- Increase quantity
- Decrease quantity
- Automatic total calculation
- Interactive buttons and UI

Django Requirements

Models

- At least **3 models**
- Related models
- ForeignKey relationships
- Database

Admin

- Register your models
- Add data using Django Admin
- Edit data using Django Admin
- Delete data using Django Admin

Views and URLs

- Create views
- Create URLs
- Use dynamic URLs where needed

Templates

Use:

- Template inheritance
- Template loops
- Conditions
- Dynamic data
- Template tags

Forms

Include:

- Checkout form
- Customer information
- Form validation

CRUD

Practice:

Create → Read → Update → Delete

Images

- Upload images
- Display images on the website

Required Cart Flow

Every project must follow this basic flow:

1. **Browse**
2. **Search / Filter**
3. **View Details**
4. **Add to Cart**
5. **View Cart**
6. **Change Quantity**
7. **Calculate Total**
8. **Checkout**
9. **Enter Name and Contact Information**
10. **Confirm Order**
11. **Save Order**
12. **Show Success Message**

Final Checklist

Before submitting, make sure your project has:

Django

- At least 3 models
- Related models
- ForeignKey relationships
- Database
- Django Admin
- Views
- URLs
- Templates
- Forms
- CRUD
- Image upload

Website

- Home page
- Navigation bar
- Search
- Filter
- Details page
- Cart
- Checkout
- Order saved in database
- Success message

JavaScript

- Search
- Filtering
- Cart interactions
- Quantity controls
- Total calculation
- Interactive UI

Design

- Modern UI
- Responsive design
- Cards
- Buttons
- Hover effects
- Animations
- Mobile-friendly layout

Final Goal

Your final project should be a **complete working Django website**, not just a few pages.

Choose **ONE** project:

E-Commerce

Sell products.

Restaurant

Sell/order meals.

Online Bookstore

Sell books.

Your final project should demonstrate:

Django + Database + Models + Templates + Forms + CRUD + HTML + CSS + JavaScript

Build a real website, not just pages.

Good , luck best wishes eng Maria